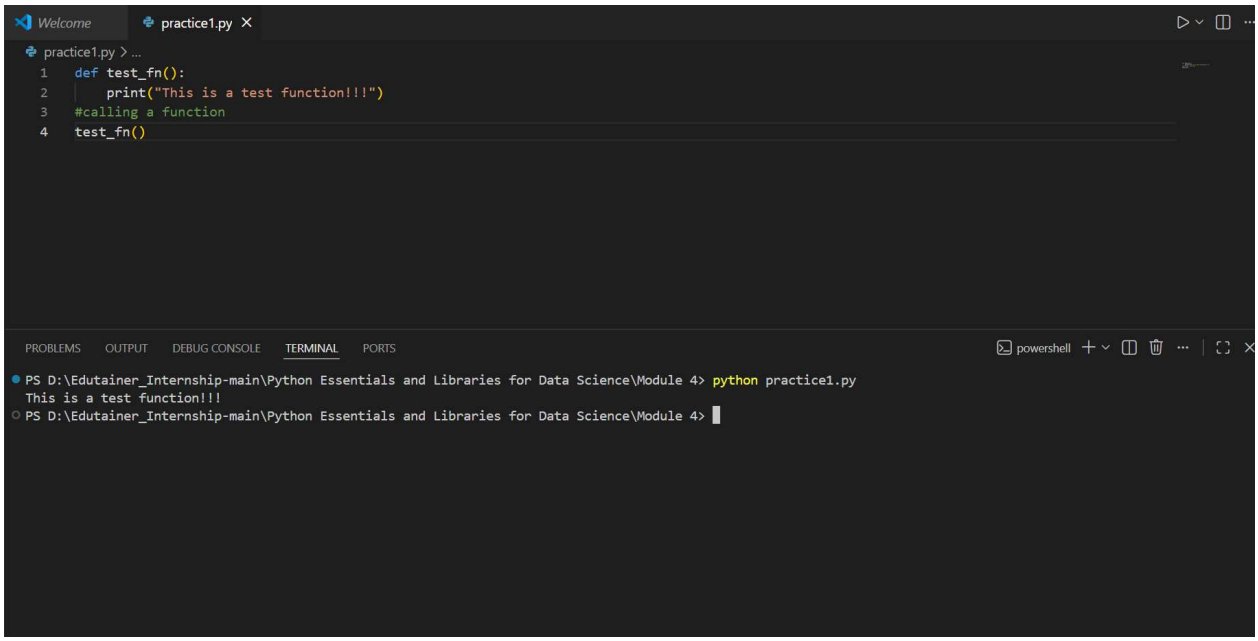
Python Functions:-

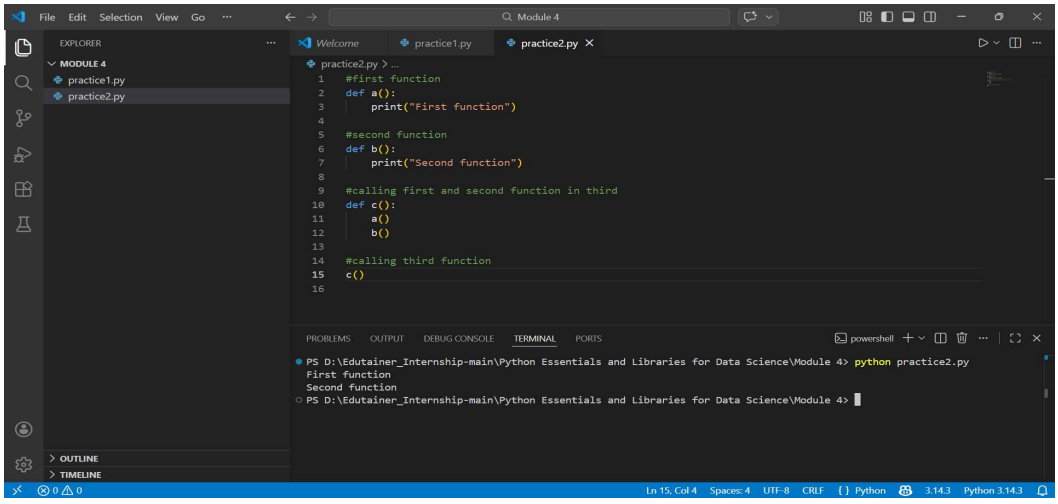
1. A function executes a set of statements as a whole.
2. A function has a name and can take parameters.
3. A function can call other functions.
4. A function created by a user is called a user defined function.
5. Functions already available in python are called library functions.

Example:-

```
def test_fun(){
   print("This is a test function.")
}
#calling a function
test_fn()
```



We call call a function inside another function.



We can also pass parameters to a function

```
def calc_disc(bill_amt):
    if bill_amt<=5000:
        print("Discount is 10%")
    else:
        print("Discount is 15%")

#calling the function
calc_disc(9000)
```

```
PS D:\Edutainer_Internship-main\Python Essentials and Libraries for Data Science\Module 4> python practice3.py
Discount is 15%
PS D:\Edutainer_Internship-main\Python Essentials and Libraries for Data Science\Module 4>
```

Here bill_amt is the parameter passed to the calc_disc function.

A function can have multiple parameters

```
def cust_record(cname, cphone, billamt):
    print("Customer name is {} , phone number is {} and  your bill amt is {}".format(cname, cphone, b

cust_record("Vamshi",1234567890,3000)

cust_record("Neelamma",1234567890,3000)

cust_record("Sonu",1234567890,3000)

cust_record("Geo",1234567890,3000)
```

```
PS D:\Edutainer_Internship-main\Python Essentials and Libraries for Data Science\Module 4> python practice4.py
Customer name is Vamshi , phone number is 1234567890 and  your bill amt is 3000
Customer name is Neelamma , phone number is 1234567890 and  your bill amt is 3000
Customer name is Sonu , phone number is 1234567890 and  your bill amt is 3000
Customer name is Geo , phone number is 1234567890 and  your bill amt is 3000
PS D:\Edutainer_Internship-main\Python Essentials and Libraries for Data Science\Module 4>
```

Passing collections as parameters and using global variable

When we have global and local variables with the same name and the local variable is inside a called function then the local variable value is displayed.

practice5.py

```python
# Passing collections as parameters and using global variable
x = [23, 34, 42, 23, 32]
t = 0
def show_biggest_no(x):
    global t
    for i in x:
        if i > t:
            t = i
    print(t)

show_biggest_no(x)
```

```
PS D:\Edutainer_Internship-main\Python Essentials and Libraries for Data Science\Module 4> python practice5.py
42
PS D:\Edutainer_Internship-main\Python Essentials and Libraries for Data Science\Module 4>
```

practice6.py

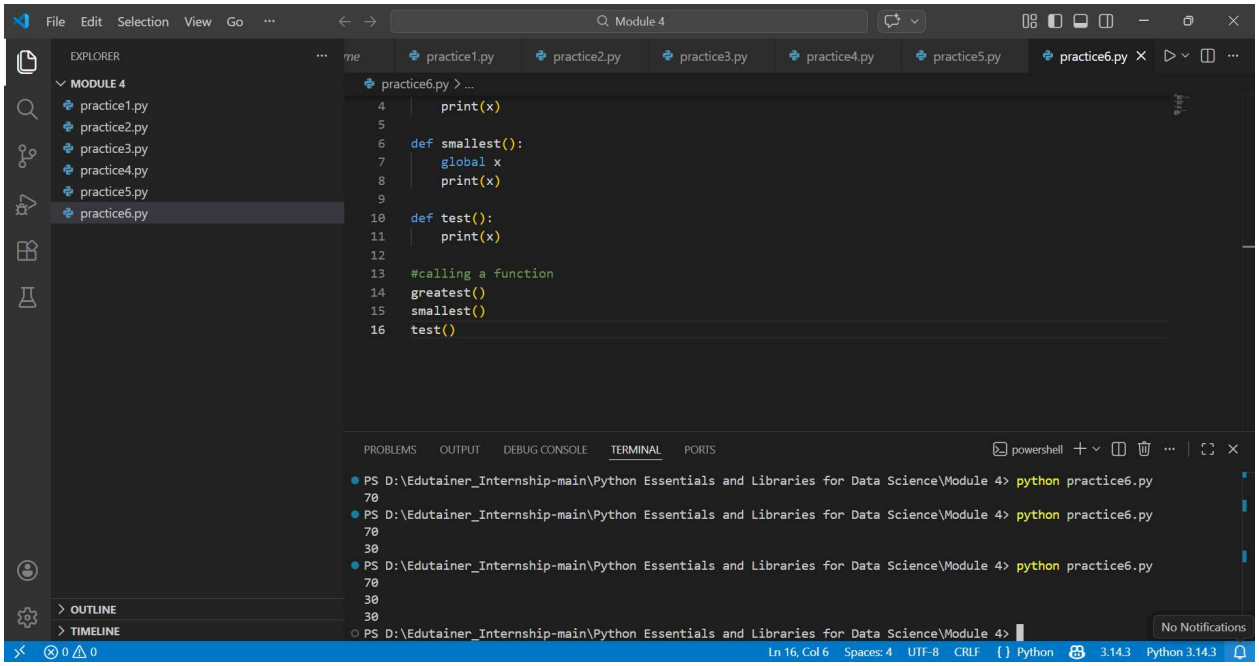
```python
x=30
def greatest():
    x=70#local variable
    print(x)

def smallest():
    global x
    print(x)

#calling a function
greatest()
smallest()
```

```
PS D:\Edutainer_Internship-main\Python Essentials and Libraries for Data Science\Module 4> python practice6.py
70
PS D:\Edutainer_Internship-main\Python Essentials and Libraries for Data Science\Module 4> python practice6.py
70
30
PS D:\Edutainer_Internship-main\Python Essentials and Libraries for Data Science\Module 4>
```

If there are no local variable then the global value is taken.


File Edit Selection View Go
Module 4
EXPLORER
MODULE 4
practice1.py
practice2.py
practice3.py
practice4.py
practice5.py
practice6.py
practice6.py > ...
print(x)
def smallest():
global x
print(x)
def test():
print(x)
#calling a function
greatest()
smallest()
test()
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
powershell
PS D:\Edutainer_Internship-main\Python Essentials and Libraries for Data Science\Module 4> python practice6.py
70
PS D:\Edutainer_Internship-main\Python Essentials and Libraries for Data Science\Module 4> python practice6.py
70
30
PS D:\Edutainer_Internship-main\Python Essentials and Libraries for Data Science\Module 4> python practice6.py
70
30
30
PS D:\Edutainer_Internship-main\Python Essentials and Libraries for Data Science\Module 4>
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